

HIRDESH SINGH

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PROFESSIONAL SUMMARY

Computer Science undergraduate with strong foundations in Java, Object-Oriented Programming, DBMS, Data Structures & Algorithms, and full-stack web development. Experienced in building responsive web applications using HTML, CSS, JavaScript, React.js, MySQL, and MongoDB. Familiar with REST APIs, Git, SDLC, and backend development concepts. Passionate about solving real-world problems through scalable and efficient software solutions.

EDUCATION

Bachelor of Technology – Computer Science & Engineering

VIT Bhopal University, Bhopal

Aug 2023 – Aug 2027 (Expected)

CGPA: 8.49/10

Class XII

Sachdeva Millennium School

Percentage: 94.6%

Class X

Sachdeva Millennium School

Percentage: 94.8%

TECHNICAL SKILLS

Programming Languages: Java, JavaScript, SQL, Python (Basics), C++ (Basics)

Frontend Technologies: HTML5, CSS3, JavaScript, React.js, Bootstrap

Backend Technologies: Node.js, Express.js, REST APIs, JDBC (Basic), Spring Boot (Basic), Spring MVC (Basic)

Databases: MySQL, MongoDB

Tools & Platforms: Git, GitHub, Postman, VS Code

Core Concepts: Object-Oriented Programming (OOP), Data Structures & Algorithms, DBMS, Operating Systems, SDLC, MVC Architecture, RESTful Web Services, Debugging & Troubleshooting

PROJECTS

Medicare – Online Disease Prediction & Doctor Appointment System | HTML, CSS, JavaScript, MongoDB

- Developed responsive user interfaces using HTML5, CSS3, and JavaScript for appointment booking and disease prediction workflows.
- Designed backend data models and implemented CRUD operations for users, doctors, and appointments using MongoDB.
- Integrated REST APIs for authentication, appointment scheduling, and role-based access control.
- Implemented secure multi-role authentication for Admin, Doctor, and User modules.
- Applied OOP principles and modular design practices to improve maintainability and scalability.

Deep-Fake Video Detection (AI + Web Integration) | Python, CNNs, RNNs, LSTMs, React.js, MongoDB

- Built an AI-powered system to detect face-swap deepfakes using a combination of CNNs, RNNs, and LSTMs for accurate classification.
- Developed an interactive React.js frontend enabling real-time video uploads, inference processing, and result visualization.
- Integrated MongoDB backend for scalable data handling and persistent storage of detection results.

- Optimized model performance using adversarial training techniques and the FaceForensics++ dataset to improve robustness.

Smart Navigation System for Visually Impaired | *Python, YOLOv8, OpenCV, Flask, pyttsx3*

- Developed a real-time AI-based assistive navigation system using YOLOv8 and OpenCV to detect objects such as stairs, vehicles, and pedestrians from live camera input.
- Designed a modular processing pipeline for object detection, decision-making, and voice-based user alerts with low-latency offline functionality.
- Integrated distance and direction estimation with a text-to-speech engine to deliver real-time voice alerts for safer navigation assistance.
- Collaborated in an Agile development workflow to integrate the computer vision pipeline, Flask backend, and web-based interface.

ACHIEVEMENTS & COMPETITIVE PROGRAMMING

- Solved 150+ DSA problems on LeetCode using Java, covering arrays, trees, graphs, and dynamic programming.
- Solved 100+ problems on GeeksforGeeks, reinforcing core algorithmic concepts and problem-solving techniques.
- Strong problem-solving skills using Java and object-oriented programming concepts.
- Participated in multiple university-level hackathons, collaborating in teams to design and prototype scalable web solutions under tight deadlines.

CERTIFICATIONS

- Introduction to Machine Learning – NPTEL
- Marketing Analytics – NPTEL
- Oracle Certified Foundations Associate – Oracle
- Generative AI Professional – Oracle
- Google IT Support Professional Certificate – Google / Coursera
- Bits and Bytes of Computer Networking – Coursera
- AWS Certification Course – Intellipaat

SOFT SKILLS

Team Collaboration • Problem Solving • Communication • Adaptability • Time Management • Attention to Detail

RELEVANT COURSEWORK

Data Structures & Algorithms • Object-Oriented Programming • Database Management Systems • Operating Systems • Software Engineering • Computer Networks • Web Technologies